Project Report: Advanced Audio Deepfake Detection using WavLM and Multi-Factor Ensemble Analysis

Prepared By: *Aryan Pandya,* B.Tech (Artificial Intelligence & Machine Learning), Indian Institute of Information Technology (IIIT), Nagpur.

---

1. Executive Summary

This project addresses the growing threat of synthetic audio media and "voice-cloning" scams proliferating on social platforms like WhatsApp. The system utilizes a hybrid approach, combining a state-of-the-art Self-Supervised Learning (SSL) foundation model (**WavLM**) with a **Multi-Factor Forensic Ensemble**. By integrating neural semantic analysis with physical and biological audio checks, the project successfully identifies deepfakes while maintaining robustness against real-world "noisy" historical recordings that often trigger false alarms in traditional detectors.

---

2. System Architecture

2.1 Neural Foundation Model

The core of the detection engine is built on the **WavLM-Base** foundation model.

- **WavLMFakeDetector:** A custom PyTorch class was implemented using the frozen microsoft/wavlm-base model as a feature extractor.
- **Classification Head:** A specialized neural head was added, consisting of a sequential stack of Linear layers (768 to 256 units), ReLU activations, Dropout (0.3) for regularization, and a final Sigmoid activation to produce a "Deepfake Penalty Score".
- **Dual-GPU Optimization:** The model was designed to support **DataParallel** training, enabling efficient distribution across multiple GPUs by detecting the data device directly from the input tensors.

2.2 Detection Geometry: Sliding Window Pipeline

To handle variable-length audio (such as 15-second WhatsApp messages), the system employs a **Sliding Window Geometry**:

- **Window Size:** 3 seconds (48,000 samples at 16kHz).
- **Step/Overlap:** 1 second, ensuring that temporal "glitches" at the edges of windows are captured by overlapping segments.
- **Inference:** Every 3-second window is passed through the model to generate a localized probability score.

---

3. Dataset and Training

- **Dataset:** The model was trained on the **"In The Wild"** audio deepfake dataset, containing over **30,000 files**.
- **Hardware:** Training was executed in a cloud environment (Kaggle) utilizing **Dual T4 GPUs**, achieving 15GB of VRAM per worker.
- **Training Performance:**
    - **Initial Loss:** The model began Epoch 1 with a Binary Cross Entropy (BCE) loss of **0.1786**, indicating rapid convergence due to the strength of the WavLM foundation.
    - **Resource Utilization:** Training sustained high GPU utilization (average 85-92%) and significant RAM pre-loading (14.3 GB) to maintain a continuous data pipeline.

---

4. Experimental Results and "Domain Gap" Analysis

During testing on real-world media, a critical "Domain Gap" was identified where the neural network alone struggled to differentiate between **acoustic noise** and **synthetic artifacts**.

- **Case Study A: Historical Real Speech (vide0_4.mp4):** A 100% real recording of PM Modi from 2016 achieved an average fake score of **0.5660**. This was a false positive caused by background newsroom noise and low-bitrate compression being interpreted as "robotic" artifacts.
- **Case Study B: AI Parody Clip (video_3.mp4):** A clean, high-quality AI parody achieved a low average score of **0.0878**, nearly passing as real because its "clean" audio lacked the noisy signatures the model associated with deepfakes.

---

5. Proposed Multi-Factor Forensic Ensemble

To solve the Domain Gap, the project transitioned from a single-factor neural model to a **Three-Judge Ensemble Architecture**:

1. **The Neural Judge (40% Weight):** The primary WavLM model analyzing semantic and deep acoustic features.
2. **The Physics Judge (30% Weight):** A spectral analysis tool checking for "High-Frequency Roll-off." Deepfakes often lack energy above 8000 Hz due to generator compression.
3. **The Biology Judge (30% Weight):** A digital signal processing (DSP) check for **Pitch Jitter**. Human vocal cords produce micro-tremors that are absent in "mathematically perfect" AI voices.

---

6. Implementation Details

- **Pre-Processing Pipeline:** Implements **Silero VAD (Voice Activity Detection)** to strip out non-speech segments and **Butterworth Bandpass Filtering** (80Hz - 7500Hz) to remove mechanical noise without destroying deepfake fingerprints.
- **Consensus Logic:** To reduce paranoia, a **Double-Lock Verdict** was implemented. A video is only flagged as a DEEPFAKE if the average score is >0.5 **and** more than 30% of the windows are suspicious.

- **Local Deployment:** The inference engine is optimized for **CPU-only** execution on consumer hardware (Lenovo IdeaPad Slim 3, 12th Gen i5), using map_location to bridge GPU-trained weights to local RAM.

---

7. Conclusion

This project successfully moves beyond basic neural classification by incorporating forensic physics and biological vocal analysis. By accounting for the "Domain Gap" through ensemble voting and robust pre-processing, the system provides a reliable tool for detecting audio deepfakes in diverse, real-world conditions.